

ECE595: Power Converters for AC Systems

Lecture 24 (Control for Power Converters & Intro to B2B Configurations)

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P2P Teaching (46.7% engagement)

Engagement & Interaction Depth

Average number of prompts per student: ~3.8

Median: 3

Range: 1 → 8+

~25–30% engaged in iterative refinement / dialogue

Short prompts (1 sentence): ~60%

Moderate prompts: ~30%

Structured / multi-part prompts: ~10%

Example (high-quality prompt):

“Explain control strategies, then compare vector control vs MPC...”

Simple prompt:

“What is a back-to-back converter?”

Basic topology (AC-DC-AC) ~95%

Applications ~80%

Control (dq, cascaded loops) ~55%

Advanced control (MPC, VSM) ~15%

Switching losses / frequency ~20%

Observation:

Students strongly converge on structure + applications

Weak penetration into control theory and design tradeoffs

Teaching what things are

Not teaching how to think

Conflicting answers?

A. Role of the DC-Link (MOST COMMON VARIATION)

Version 1 (simplified)

“DC-link is just a capacitor that stores energy”

Version 2 (more correct)

“DC-link decouples systems and balances power dynamically”

B. Control Responsibility Split

Version 1 (correct)

Grid-side → DC-link voltage

Load-side → output control

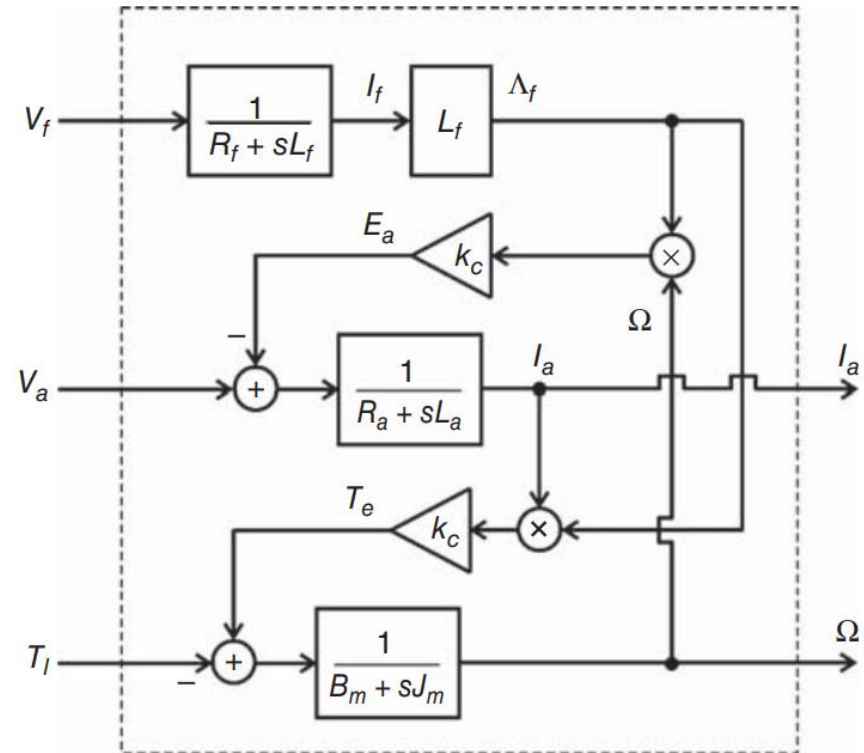
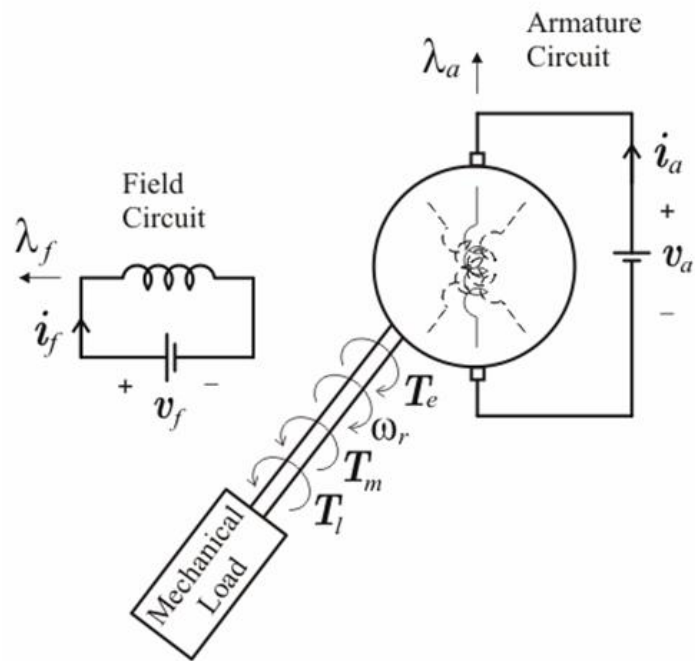
Version 2 (less clear)

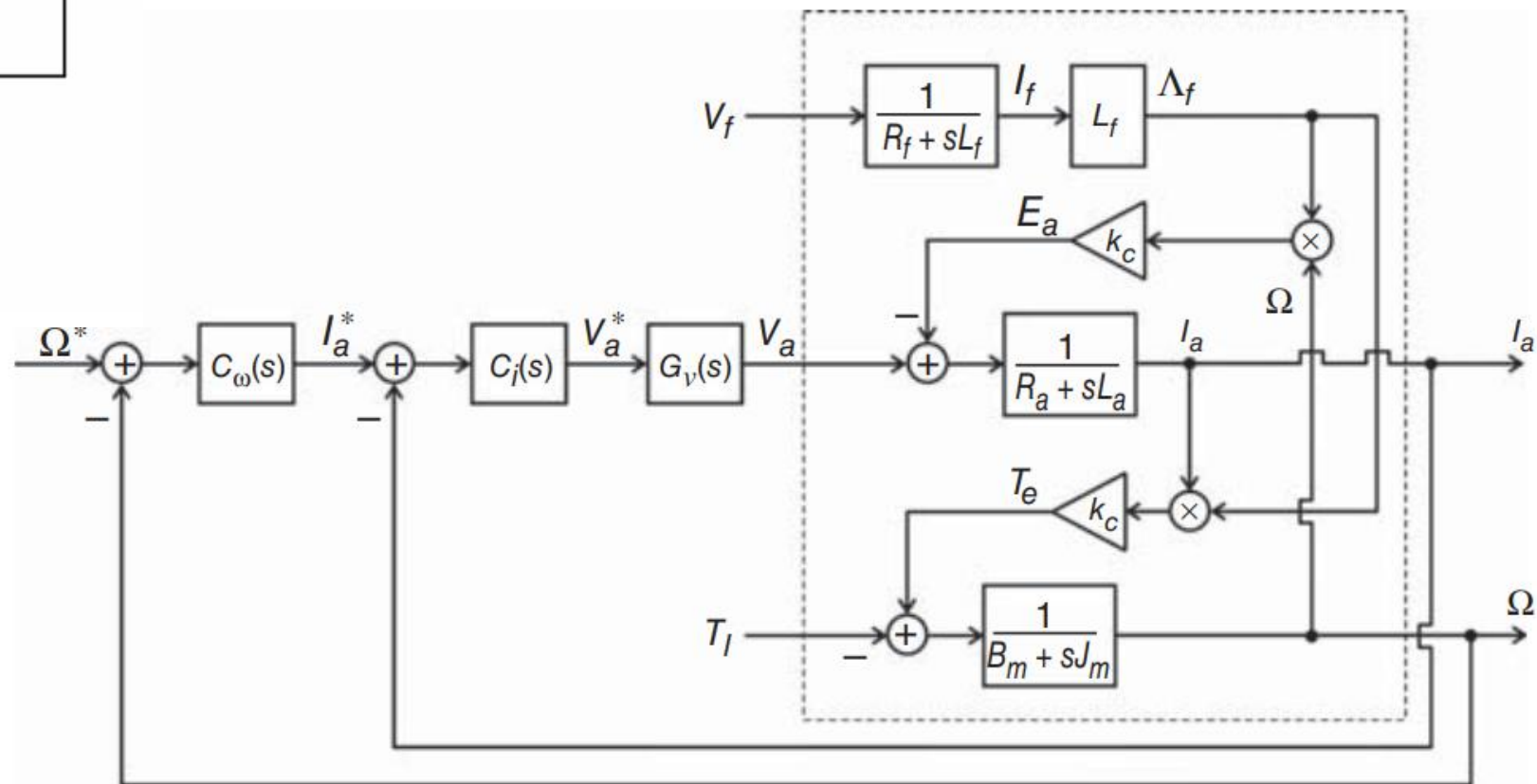
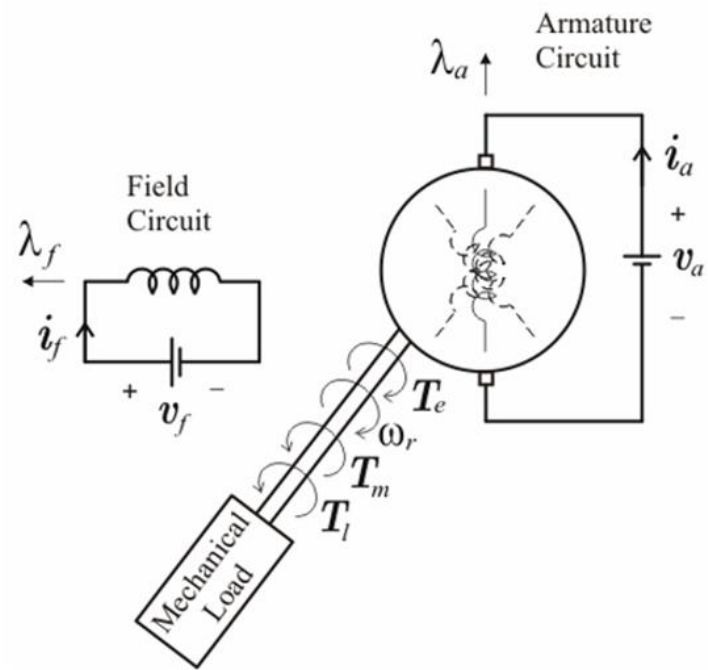
“Both converters control everything”

Version 3 (confused)

No clear separation of roles

$$\frac{d}{dt} \begin{bmatrix} i_a \\ \omega \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -k_c \lambda_f/L_a \\ k_c \lambda_f/J_m & -B_m/J_m \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a & 0 \\ 0 & -1/J_m \end{bmatrix} \begin{bmatrix} v_a \\ T_l \end{bmatrix}$$





LINEAR CONTROL – ac VARIABLE

Resonant PI type of controller

$$C(s) = \frac{k_a s^2 + k_b s + k_c}{(s^2 + \omega^2)}$$

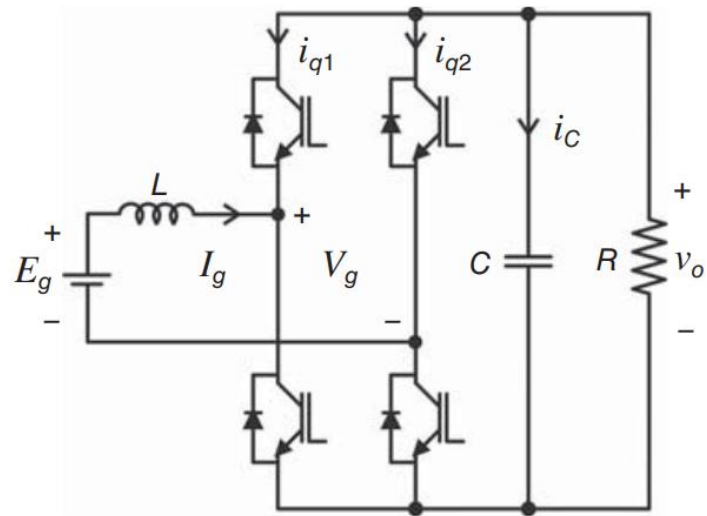
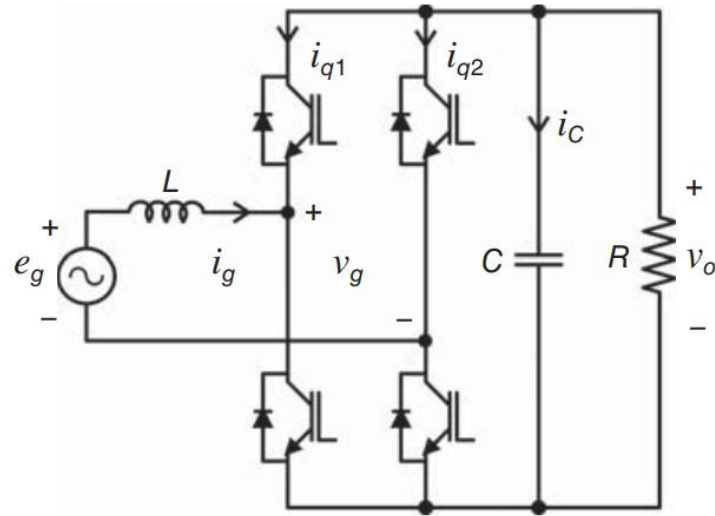
state space representation of this controller

$$\dot{x}_1 = x_2 + k_b e$$

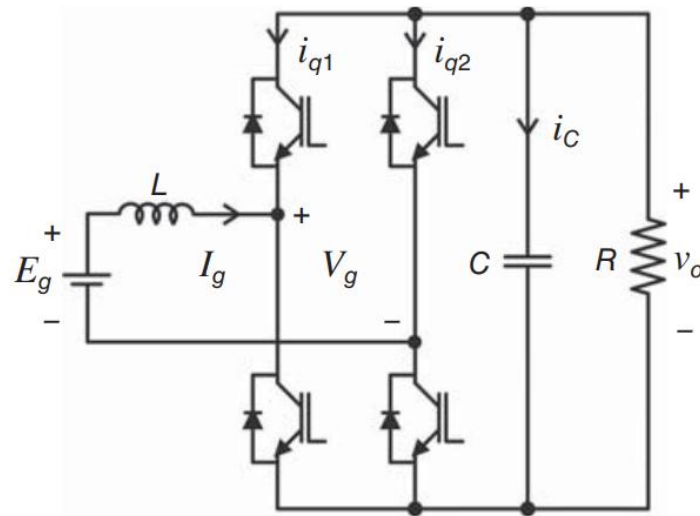
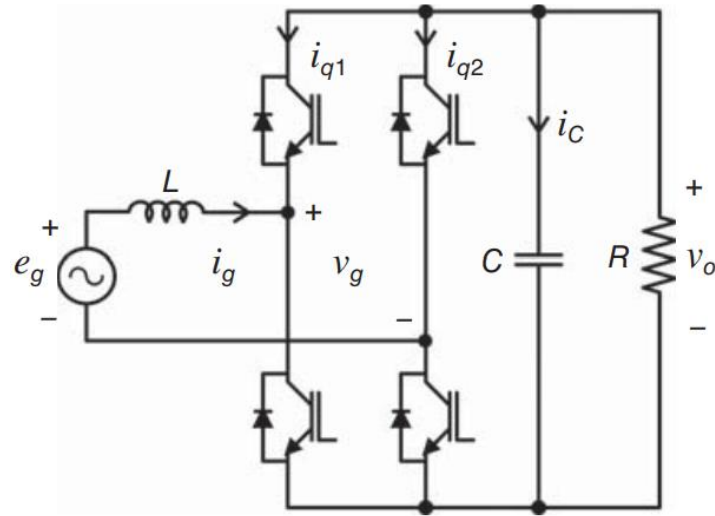
$$\dot{x}_2 = -\omega^2 x_1 + k_d e$$

$$y = x_1 + k_a e$$

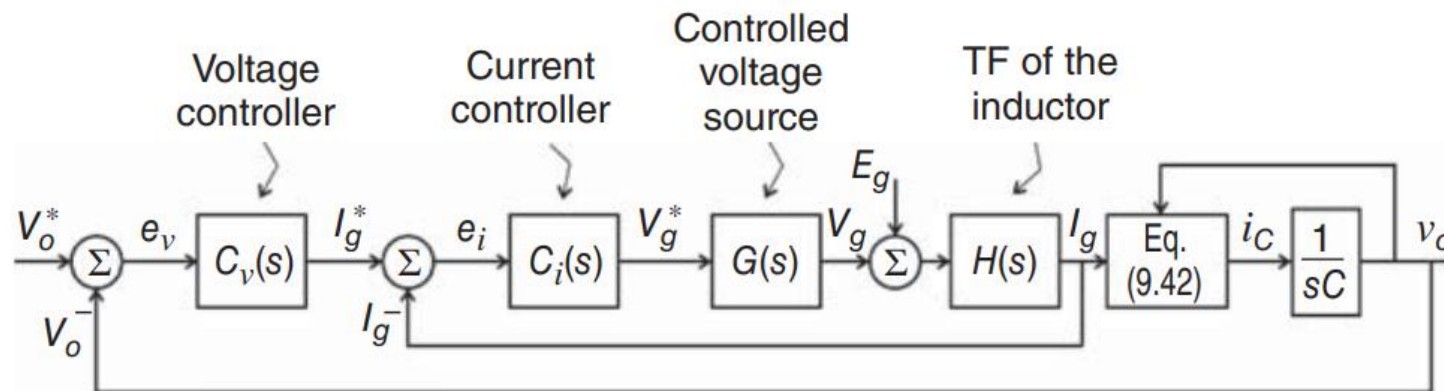
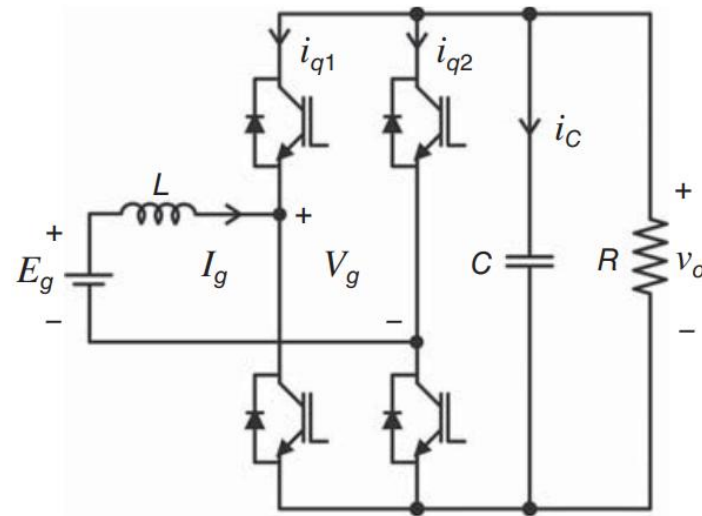
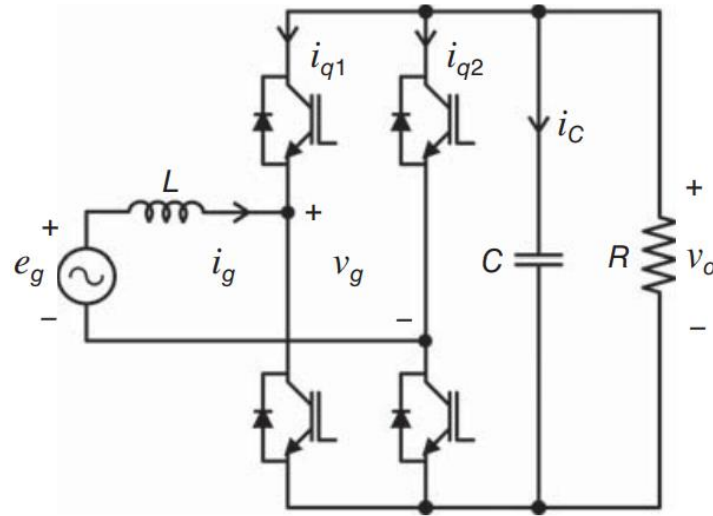
CASCADE CONTROL STRATEGIES



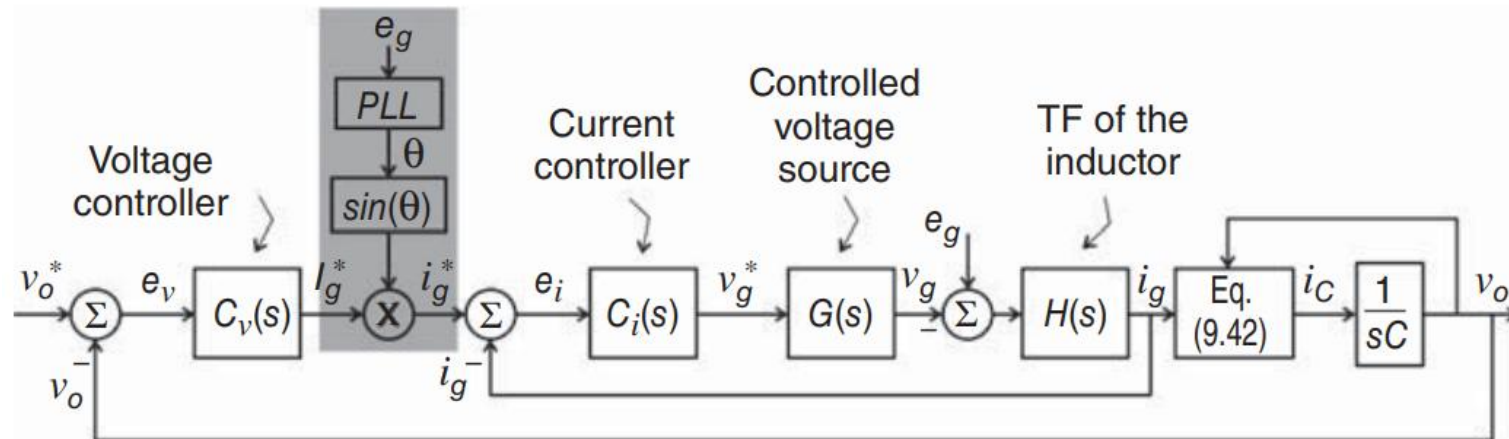
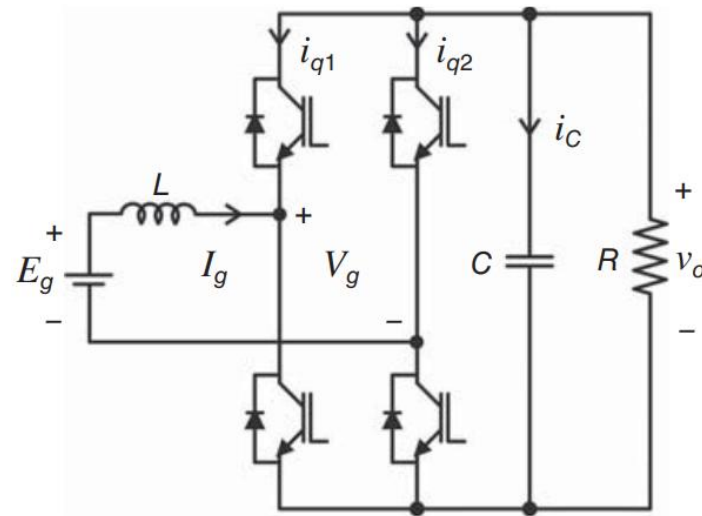
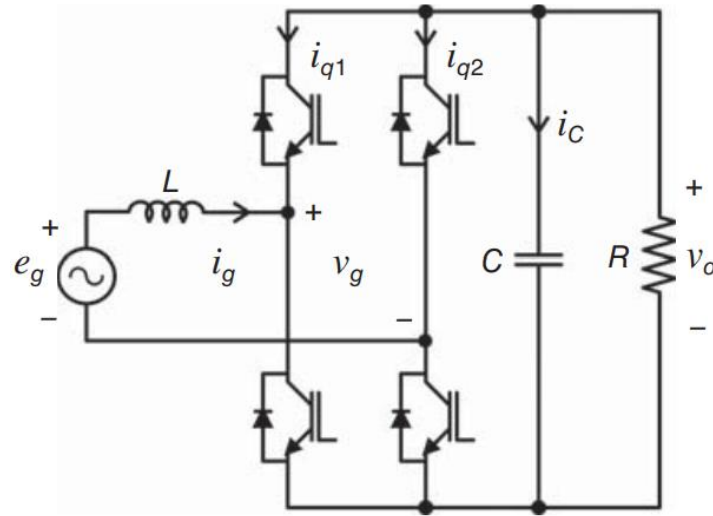
CASCADE CONTROL STRATEGIES



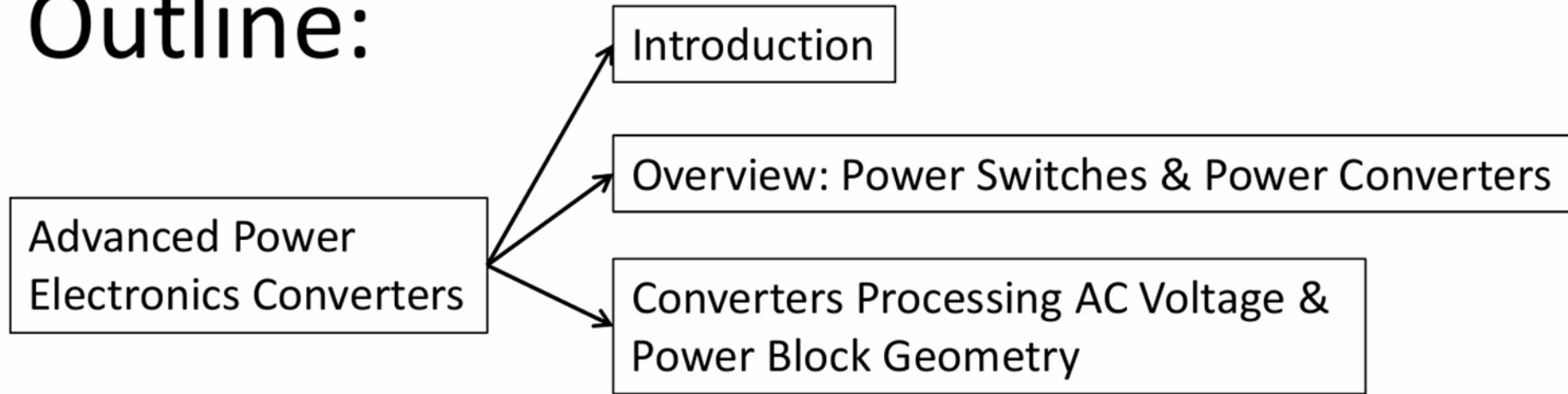
CASCADE CONTROL STRATEGIES



CASCADE CONTROL STRATEGIES



Outline:



Multi-level Converters:

- Neutral-Point-Clamped Configuration
- Cascade Configuration
- Flying-Capacitor Configuration
- Non-Conventional Topologies

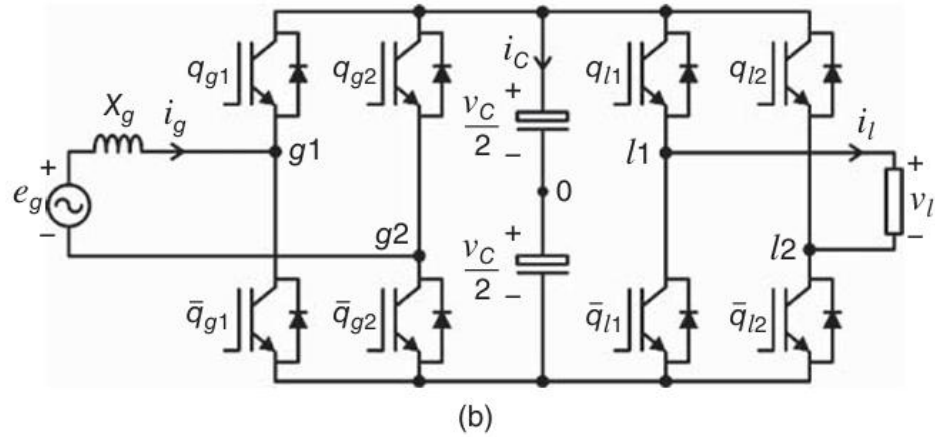
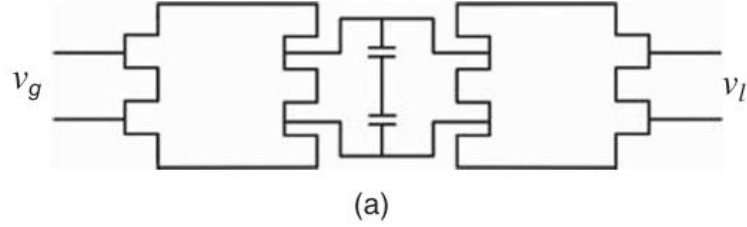
Optimized PWM Approach

Control Strategies for
Power Converters

Back-to-back Converters:

- Single-phase system
- Three-phase system
- Non-Conventional Topologies

Modeling



$$v_{g10} = (2q_{g1} - 1) \frac{v_C}{2}$$

$$v_{g20} = (2q_{g2} - 1) \frac{v_C}{2}$$

$$v_{l10} = (2q_{l1} - 1) \frac{v_C}{2}$$

$$v_{l20} = (2q_{l2} - 1) \frac{v_C}{2}$$

$$v_g = v_{g10} - v_{g20}$$

$$v_g = e_g - r_g i_g - l_g \frac{di_g}{dt}$$

$$v_c = \frac{1}{C} \int [(q_{g1} - q_{g2})i_g + (q_{l2} - q_{l1})i_l] dt$$

PWM

$$v_{g10}^* = v_g^* + v_{gh}^*$$

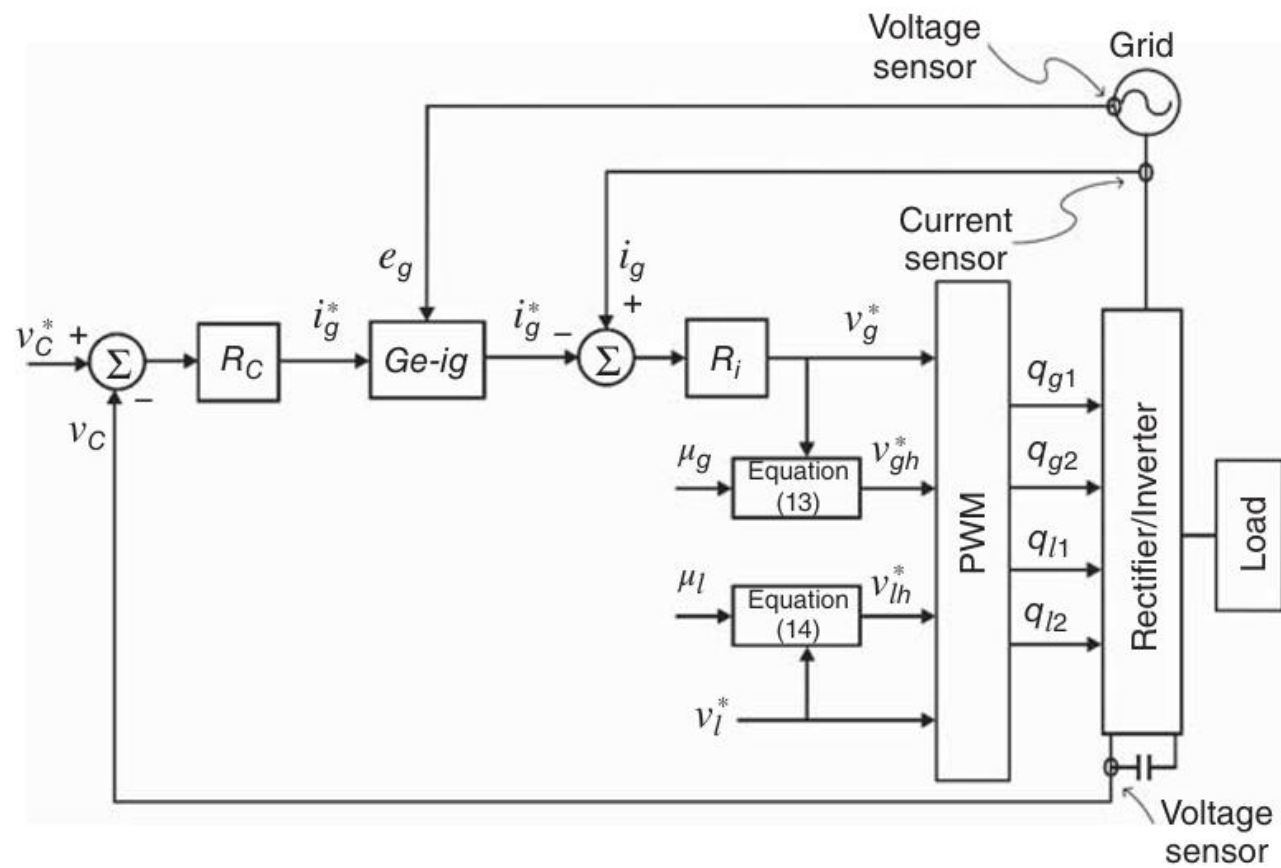
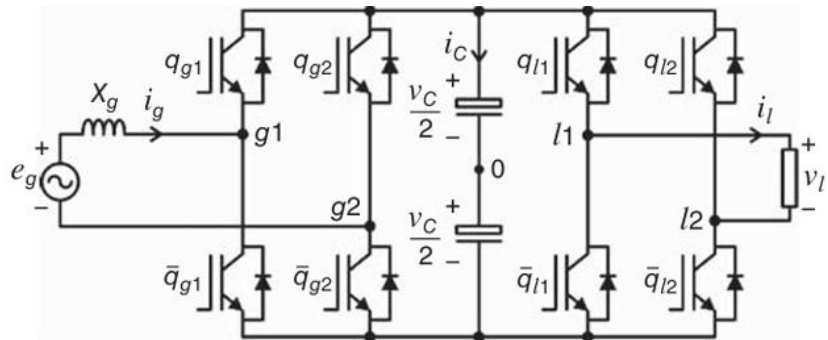
$$v_{g20}^* = v_{gh}^*$$

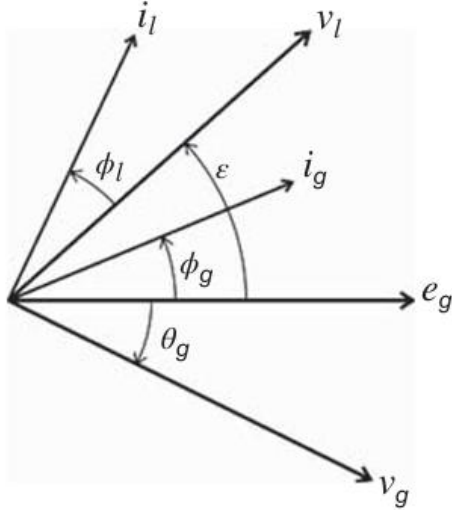
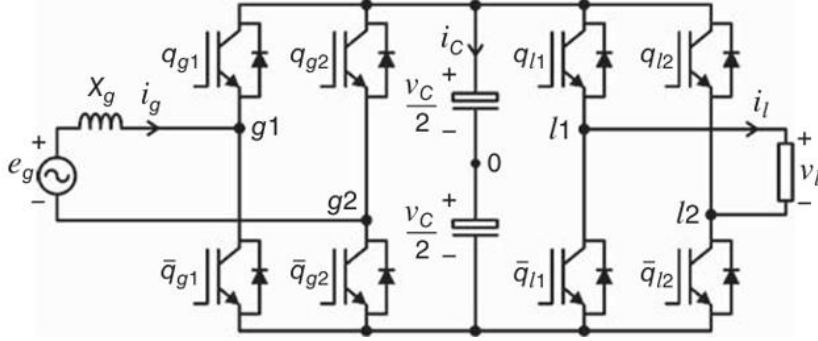
$$v_{l10}^* = v_l^* + v_{lh}^*$$

$$v_{l20}^* = v_{lh}^*$$

$$v_{gh}^* = v_C \left(\mu_g - \frac{1}{2} \right) + (\mu_g - 1)V_{g\max}^* - \mu_g V_{g\min}^*$$

$$v_{lh}^* = v_C \left(\mu_l - \frac{1}{2} \right) + (\mu_l - 1)V_{l\max}^* - \mu_l V_{l\min}^*$$





$$p_c(t) = p_g(t) - p_l(t)$$

$$p_c(t) = \left[\frac{V_g I_g}{2} \cos(\theta_g + \Phi_g) - \frac{V_l I_l}{2} \cos(\Phi_l) \right] + \left[\frac{V_g I_g}{2} \cos(2\omega_g t - \theta_g + \Phi_g) - \frac{V_l I_l}{2} \cos(2\omega_l t + 2\epsilon + \Phi_l) \right]$$

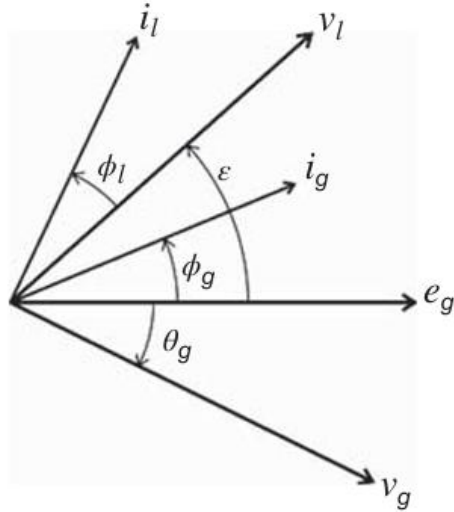
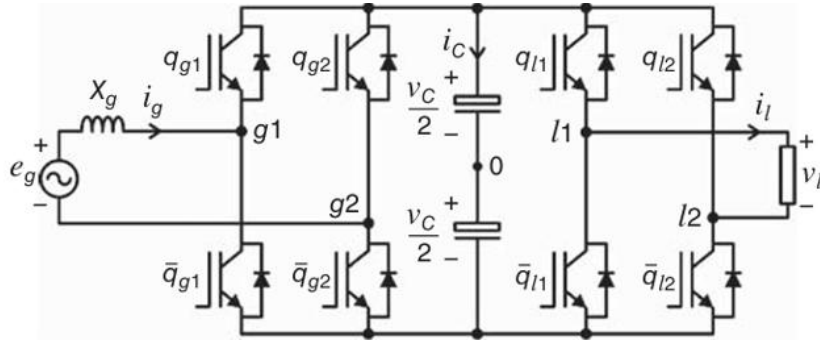
$$V_g I_g \cos(\theta_g + \Phi_g) = V_l I_l \cos(\Phi_l)$$

$$p_c(t) = \frac{V_g I_g}{2} \cos(2\omega_g t - \theta_g + \Phi_g) - \frac{V_l I_l}{2} \cos(2\omega_l t + 2\epsilon + \Phi_l)$$

$$p_c(t) = N_l \frac{\cos(\Phi_l)}{\cos(\theta_g + \Phi_g)} \cos(2\omega_g t - \theta_g + \Phi_g) - N_l \cos(2\omega_l t + 2\epsilon + \Phi_l)$$

where $N_l = V_l I_l / 2$.

$$p_c(t) = v_c i_c = C v_c \frac{dv_c}{dt}$$



$$p_c(t) = v_c i_c = C v_c \frac{dv_c}{dt}$$

$$v_c \frac{dv_c}{dt} = \frac{N_l}{C} \left[\frac{\cos(\Phi_l)}{\cos(\theta_g + \Phi_g)} \cos(2\omega_g t - \theta_g + \Phi_g) - \cos(2\omega_l t + 2\epsilon + \Phi_l) \right]$$

Applying an integral at both sides of equation

$$v_c^2 = N_l \left[\frac{\cos(\Phi_l)}{\cos(\theta_g + \Phi_g)} X_{cg} \sin(2\omega_g t - \theta_g + \Phi_g) - X_{cl} \sin(2\omega_l t + 2\epsilon + \Phi_l) \right] + \bar{v}_c$$

$$v_c = \sqrt{\bar{v}_c + \tilde{v}_c}$$

$$\tilde{v}_c = N_l X_{cl} \left[\frac{\cos(\Phi_l)}{\cos(\theta_g + \Phi_g)} \frac{f_l}{f_g} \sin(2\omega_g t - \theta_g + \Phi_g) - \sin(2\omega_l t + 2\epsilon + \Phi_l) \right]$$

